

The Right-Triangle Solver's Guide

Every problem in this chapter is one of six situations. Name the situation, and the tool picks itself.

The situation	The tool	How it works
Two sides known, want the third	Pythagorean Theorem	$a^2 + b^2 = c^2$ — c is ALWAYS the hypotenuse
45°-45°-90° or 30°-60°-90° pattern	special right triangle ratios	exact answers with $\sqrt{2}$ or $\sqrt{3}$ — no calculator
One acute angle + one side, want a side	sin / cos / tan	label opp-adj-hyp FROM THE ANGLE, then pick the ratio using what you know and what you want
Two sides known, want an angle	$\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$	inverse ratios turn a side ratio back into the angle
Story problem with looking up/down	angle of elevation / depression	draw the horizontal first; the angle sits at the observer; it is almost always a tangent equation
Want EVERYTHING (all sides, all angles)	solve the right triangle	Pythagorean for the third side, one inverse ratio for an angle, subtract from 90° for the other

The Two Mistakes That Cause Most Lost Points

- Using a leg as c. In $a^2 + b^2 = c^2$, the letter c is not 'the third side' — it is the HYPOTENUSE, every time. When a hypotenuse and a leg are given, SUBTRACT.
- Labeling opp/adj from the wrong angle. Opposite and adjacent are defined relative to the angle you are using. Re-label the sides every time the angle changes.

Exact Values Worth Knowing

These come straight from the two special right triangles — no memorizing needed if you can redraw them.

	30°	45°	60°
sin	1/2	$\sqrt{2}/2$	$\sqrt{3}/2$
cos	$\sqrt{3}/2$	$\sqrt{2}/2$	1/2
tan	$\sqrt{3}/3$	1	$\sqrt{3}$

Radical Skills on One Line Each

- Simplify: pull out the largest perfect square — $\sqrt{72} = \sqrt{(36 \cdot 2)} = 6\sqrt{2}$.
- Rationalize: multiply top and bottom by the radical — $14/\sqrt{2} = 14\sqrt{2}/2 = 7\sqrt{2}$.
- Special-right shortcuts: leg → hypotenuse is MULTIPLY by $\sqrt{2}$ (or 2); hypotenuse → leg is DIVIDE.

Elevation and Depression Without Tears

- Draw the horizontal line FIRST, at the observer's eye. Elevation opens upward from it; depression opens downward.
- The depression angle at the top equals the elevation angle at the bottom — alternate interior angles between parallel horizontals.
- The height and the horizontal distance are the two legs, so the equation is almost always $\tan(\text{angle}) = \text{height} / \text{horizontal}$.
- Two-observation problems (two boats, two positions): solve each triangle separately, then subtract the distances.

Classifying Triangles by Sides (9.1)

- Reality check first: the two smaller sides must ADD to more than the largest, or there is no triangle at all.
- Then compare c^2 with $a^2 + b^2$ (c = largest): equal → right · c^2 bigger → obtuse · c^2 smaller → acute.

Name: _____

Which Tool? Practice

Name the tool — Pythagorean Theorem, special right triangle, trig ratio (say which), inverse trig ratio, or elevation/depression setup. Do not solve.

1. A right triangle has legs 9 and 13; find the hypotenuse.
2. A right triangle has a 45° angle and a leg of 6; find the hypotenuse exactly.
3. A right triangle has a 28° angle with adjacent side 40; find the opposite side.
4. A right triangle has legs 7 and 12; find the smaller acute angle.
5. An equilateral triangle has side 16; find its height exactly.
6. From 500 ft away, the top of a tower sits at 36° above horizontal; find the height.
7. A right triangle has hypotenuse 41 and leg 9; find the other leg.
8. A right triangle has a 63° angle and hypotenuse 25; find the side opposite the angle.
9. A ski slope drops at an 18° angle of depression over a 900-ft horizontal run; find the drop.
10. A right triangle has hypotenuse 30 and leg 24; find the angle opposite that leg.
11. A square has diagonal 22; find its side exactly.
12. Do the sides 6, 9, 16 make a triangle? If so, what kind?
13. A right triangle has a 52° angle with opposite side 31; find the hypotenuse.
14. Classify the triangle with sides 10, 12, 15.
15. Find every side and angle of a right triangle with legs 5 and 12.